



Submitted By:

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Assignment:

Lab No. 1

Submitted To:

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For the fulfillment of,

Object-Oriented Programming Lab

Department of Computer Engineering

University of Engineering and Technology, Lahore

Task No. 1:

Change the Factorial code discussed in class so that it shows proper messaging in case of negative numbers.

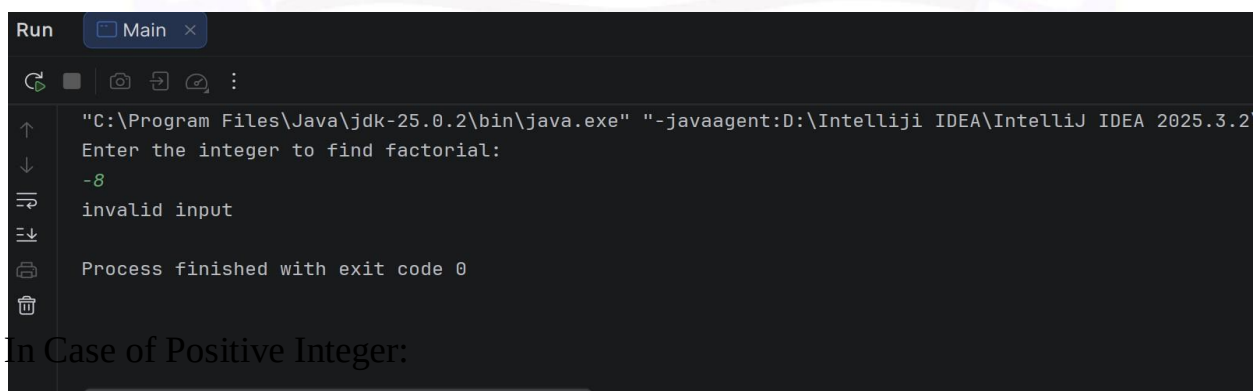
Solution:

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        int fact=1;
        Scanner scan=new Scanner(System.in);
        System.out.println("Enter the integer to find factorial:");
        int num=scan.nextInt();
        if (num>=0) {

            for (int i = 1; i <= num; i++) {
                fact = fact * i;
            }
            System.out.println("The factorial of " + num + " is " + fact);
        }
        else{
            System.out.println("invalid input:");
        }
    }
}
```

Output:

In Case of Negative Integer:



The screenshot shows a 'Run' window in an IDE. The command line at the top is: `"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA\IntelliJ IDEA 2025.3.2"`. The standard output shows the prompt "Enter the integer to find factorial:", followed by the user input `-8` (highlighted in green), and then the program output `invalid input`. At the bottom, it states "Process finished with exit code 0".

In Case of Positive Integer:



The screenshot shows a 'Run' window in an IDE. The command line at the top is: `"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA\IntelliJ IDEA 2025.3.2"`. The standard output shows the prompt "Enter the integer to find factorial:", followed by the user input `6` (highlighted in green), and then the program output `The factorial of 6 is 720`. At the bottom, it states "Process finished with exit code 0".

Task No. 2:

Change the Series Generator such that it can generate both increasing series or decreasing series depending on the starting and ending values.

Solution:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter starting value: ");
        int start = sc.nextInt();
        System.out.println("Enter ending value: ");
        int end = sc.nextInt();
        System.out.println("Enter step value: ");
        int step = sc.nextInt();
        if (start < end) {
            // Increasing series
            for (int i = start; i <= end; i += step) {
                System.out.print(i + " ");
            }
        } else if (start > end) {
            // Decreasing series
            for (int i = start; i >= end; i -= step) {
                System.out.print(i + " ");
            }
        } else {
            System.out.println("Start and end values are equal: " + start);
        }
    }
}
```

Output:

A screenshot of a terminal window showing the execution of a Java program. The program prompts the user for three inputs: starting value, ending value, and step value. The user enters 1, 20, and 2 respectively. The program then prints the resulting increasing series: 1 3 5 7 9 11 13 15 17 19. The terminal output is as follows:

```
Enter starting value:
1
Enter ending value:
20
Enter step value:
2
1 3 5 7 9 11 13 15 17 19
Process finished with exit code 0
```

Task No. 3:

Write a Program that takes year as an input from user and determine if it's a leap year or not.

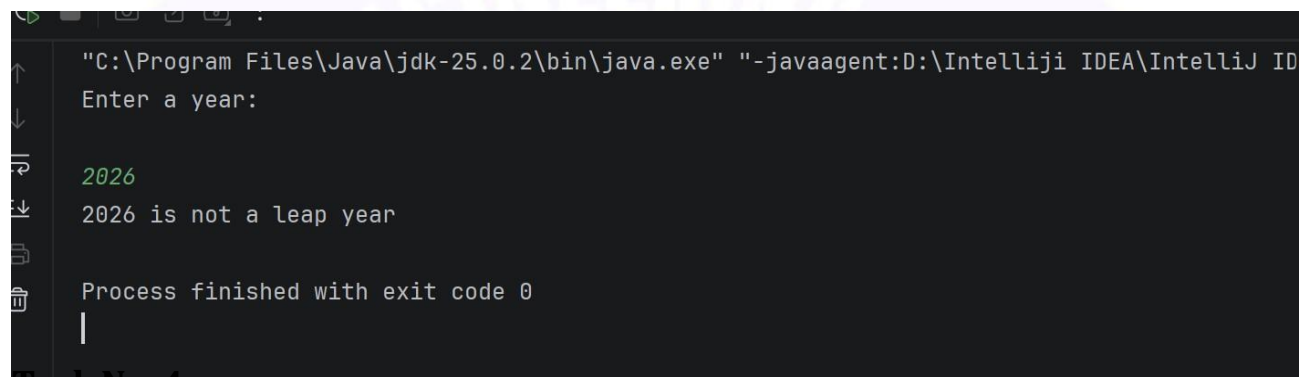
Solution:

```

import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner scan=new Scanner(System.in);
        System.out.println("Enter a Year :");
        int year =scan.nextInt();
        if ((year % 4 == 0 && year % 100 != 0 || year % 400 == 0) {
            System.out.println( year+" is a Leap Year");}
        else {
            System.out.println( year+" is not a Leap Year");
        }
    }
}

```

Output:



```

"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA\IntelliJ ID
Enter a year:
2026
2026 is not a leap year
Process finished with exit code 0

```

Task No. 4:

You are interested in calculating your electricity bill of your home. You know the unit consumption of your home. Billing criteria for electric company is that it charges 5 Rs/ unit for first 100 units, next 100 units are charged at 7 Rs/unit and above that charge is 15 Rs/unit. As input, you get total Units consumed and generate bill.

Solution:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the number of Units consumed:");
        int unit = sc.nextInt();
        System.out.println("Units consumed :" + unit);
        int Bill = 0, price = 0;
        if (unit <= 100) {
            price += unit * 5;

```

```

        Bill += price;
        System.out.println("Bill: "+Bill);
    }
    else if (unit <= 200) {
        price += 100 * 5;
        price += (unit - 100) * 7;
        Bill += price;
        System.out.println("Bill: "+Bill);
    }
    else if (unit > 200) {
        price += 100 * 5;
        price += 100 * 7;
        price += (unit - 200) * 15;
        Bill += price;
        System.out.println("Bill: "+Bill);
    }
}
}

```

Output:



```

"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA\Inte
Enter the number of Units consumed:
450
Units consumed :450
Bill:4950

Process finished with exit code 0

```

Task No. 5:

Write a program that takes an integer n from command-line arguments and uses a while loop to compute the number of times you need to divide n by 2 until it is strictly less than 1. Print the error message “Illegal input” if n is negative.

Solution:

```

import java.util.Scanner;

public class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

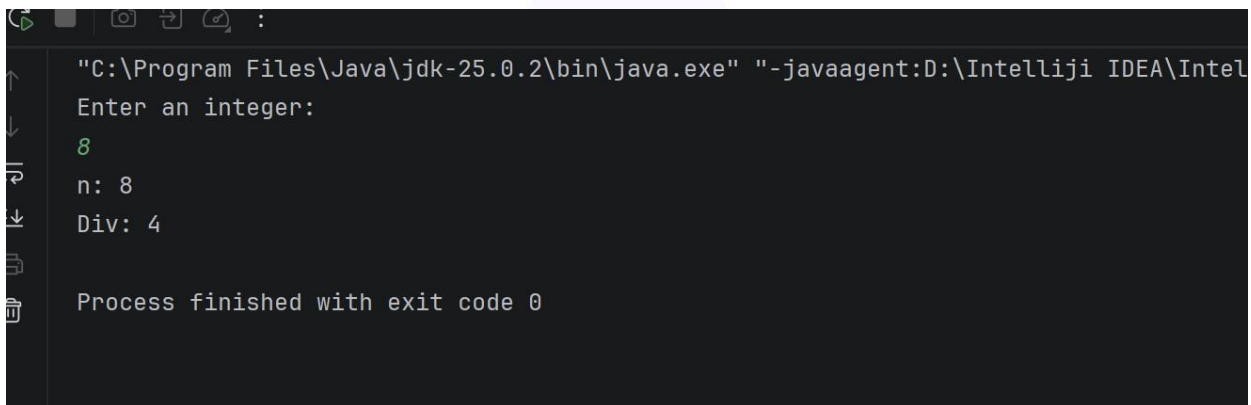
```

```

        System.out.println("Enter an integer:");
        int num = sc.nextInt();
        System.out.println("n: " + num);
        int count = 0;
        while (num >= 1) {
            num /= 2;
            count++;
        }
        System.out.println("Div: " + count);
    }
}

```

Output:



```

"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA\Intel
Enter an integer:
8
n: 8
Div: 4

Process finished with exit code 0

```

Task No. 6:

Write a code that takes a number value from user and displays the inverse of it. Number can be of any length.

Solution:

```

import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a number:");
        int num= sc.nextInt();
        System.out.println("Number: " + num);
        int reversed = 0;
        while(num!=0) {
            int digit = num % 10;
            reversed= reversed*10 + digit;
            num /= 10;
        }
        System.out.println("Reversed: "+reversed);
    }
}

```

Output:

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA\
Enter a number:
2026
Number: 2026
Reversed: 6202

Process finished with exit code 0
|
```

